

Datasheet: AC Monitor CAN

Model: ACMCANDN



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Scope

This data sheet provides technical information regards the Enatel AC monitor (ACM) model number ACMCANDN.

Product Definition

Hardware Version: Rev 5

Firmware Version: 6

Compatibility

Controller	EM4x <i>Note: EM4x firmware version: 11.0 and greater required</i>
Number of modules supported	6 ACMs per EM4x energy manager controller <i>Note: the total number of CAN bus devices that can connect to an EM4x is 32</i>
Current Transformers	
CT1P060A	Single Phase PCB 60A/Ph
CT3P060A	Three phase PCB 60A/Ph
CT3P150A	Three phase PCB 150A/Ph
CT100A33	Single phase Clip-on 100A/Ph

DC Input Voltage

Input	12V (Supplied via EM4x CAN bus) 94mA
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Mechanical

Dimensions	W: 17mm (0.67in) H: 90mm [120 mm with connectors] (3.54in [4.72in with connectors]) D: 74mm (2.91 in)
Weight	80g (0.18lb)
Mounting options	DIN Rail

Front Panel

Status LEDs	
Green	Powered on
Red	Alarm
L1	Line 1 Present
L2	Line 2 Present
L3	Line 3 Present
	<i>Note: Present is Phase-to-Neutral voltage above 100.0V</i>

Environmental Requirements

Ambient Temperature

Rated Operating Range:	- 10°C to + 55°C (14°F to 131°F)
Storage Range:	- 25°C to + 75°C (-13°F to 167°F)

Humidity

Nominal:	50% RH
Range:	< 85% RH (non-condensing)

Altitude:	< 2000m
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Note: Reduce max ambient temperature by 4°C per 1000m above sea level

Communications

System communications	CAN (Molex) <i>Note: Internal Comms only</i>
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Software upgrades	Via EM4x
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Alarms

Alarm List	No CAN Address.
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ACM Input

3 phase voltage inputs

Configurations supported.	3 Phase Star with Neutral 3 Phase Star No Neutral 3 Phase Delta 3 Independent Single-Phase Input
Cable size	0.5mm ² – 1mm ²
Measurement range	100-277V (Phase to Neutral, Nominal line voltage) 200-480V (Phase to Phase, Nominal line voltage)
Frequency Range	35-75Hz (Nominal: 50Hz)

Neutral inputs

Number	2
Cable size	0.5mm ² – 1mm ²

Earth input.

Number	1
Cable size	0.5mm ² – 1mm ²

Current transformer

Inputs	3
Cable size	0.5mm ² – 1mm ²
Input range	0-100mA

CAN

Input	1 (Connector: Molex)
Output	1 (Connector: Molex)

ACM Process and Functions

Surge Level detection	1.0 kV to 2.0 kV
	2.0 kV to 3.0 kV
	3.0 kV to 4.0 kV
	4.0 kV to 5.0 kV
	5.0 kV to 6.0 kV
	>6kV

Compliances

Safety:	IEC 61010-1
	IEC 61010-2-30

EMC – Immunity

Electrostatic discharge:	EN 61000-4-2
Radiated radio frequency:	EN 61000-4-3
Electrical fast transients:	EN 61000-4-4
Surge:	EN 61000-4-5
Conducted radio frequency:	EN 61000-4-6
Dips, interruptions & variations:	EN 61000-4-11

EMC – Emissions

Conducted emissions (AC):	EN 55032 / CISPR 32 (Class B)
Radiated emissions:	EN 55035 / CISPR 35 (Class B)
Harmonics:	EN 61000-3-2
Fluctuations and flicker:	EN 61000-3-3

Certifications

Europe	CE-mark
Australia/New Zealand	RCM